Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. All experiments were run on a single GPU with 24 GB of memory.

2 Results

Figure 1 shows the test accuracy of BERT-base by dataset.

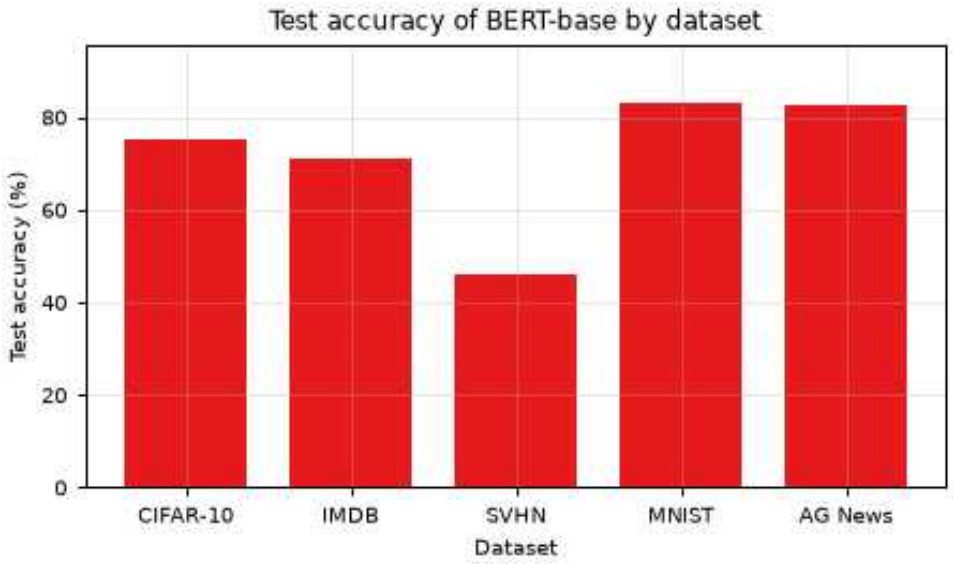


Figure 1: Test accuracy of BERT-base by dataset for the models in this experiment.

BERT-base scores 25.1 points higher on IMDB than on SVHN. On IMDB, BERT-base obtains a test accuracy of 71.0% (Figure 1). BERT-base achieves 83.0% test accuracy on MNIST. BERT-base reaches 35.0% test accuracy on AG News (Figure 1).

Figure 2 shows the validation accuracy during training.

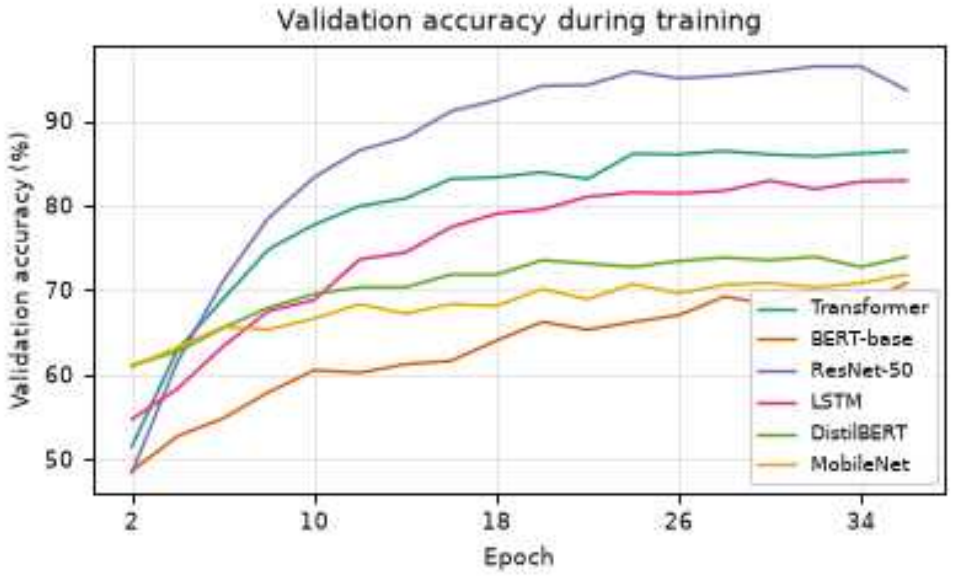


Figure 2: Validation accuracy during training for the models in this experiment.

At epoch 22, DistilBERT achieves 78.4% validation accuracy. At epoch 30, Transformer is about 18 points better than BERT-base. LSTM outperforms MobileNet by 11 percentage points at epoch 24 (Figure 2).

3 Discussion

On IMDB, Naive Bayes obtains an accuracy of 61.4%. We note that the training was stable for all models.